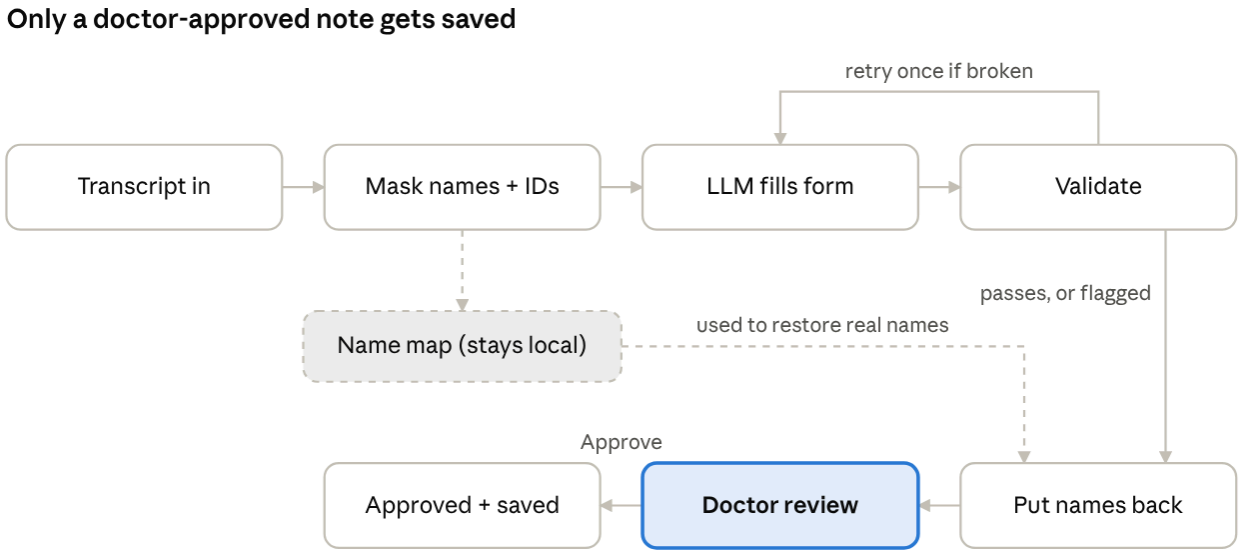
MediFlow AI — Module 1 Architecture (Scribe)

Module 1 turns a consultation transcript into a structured clinical note that only becomes official after a doctor approves it. Each section below can be handed to a coding agent as-is.

Flow at a glance



Module 1 flow · 7 components

The top row runs automatically; the bottom row is where the doctor takes over. The name map never leaves our system: it is only used to put real names back after the LLM has finished.

Components

Seven components run in a fixed order. Each one has one job, one input and one output.

C1. Transcript input

- **Job:** load a consultation for processing.
- **Input:** a transcript file from `data/transcripts/` (10 synthetic files for the prototype), picked from a dropdown on the screen.
- **Output:** the transcript text plus its header: transcript ID, patient ID, patient name, phone, NHIS number, consultation date, clinic.

C2. Masking (hide identity)

- **Job:** remove anything that identifies the patient before text leaves our system.
- **What gets masked:** patient name, any other person's name, phone numbers, NHIS numbers, patient ID.
- **How:** names and IDs come from the transcript header, so we replace those exact strings. Phone numbers (`0XX XXX XXXX` or `+233...`) and NHIS numbers (`NHIS-XXXXXXXX`) are also caught by pattern matching in case they are spoken in the dialogue.
- **Replacements:** `[PATIENT_1]`, `[PERSON_1]`, `[PHONE_1]`, `[NHIS_1]`, `[PATIENT_ID_1]`.
- **Output:** the masked transcript, plus a **name map** (placeholder → real value). The name map stays in our system and is never sent to the LLM.

C3. LLM extraction

- **Job:** read the masked transcript and fill the form (see *The form*).
- **Input:** masked transcript, consultation date, the prompt (see *The LLM prompt*).
- **How:** call Claude through the provider layer with temperature 0 and the API's tool-use / JSON schema feature, so the answer is forced into the form's shape.
- **Output:** a filled form as JSON, still containing placeholders.

C4. Validation

- **Job:** check the LLM's answer before anyone sees it.
- **How:** check against the form schema and the rules in *Guardrails*.
- **Output:** either a valid form, or a form with a list of flagged fields.

C5. Unmasking (put real names back)

- **Job:** replace every placeholder with its real value using the name map.
- **Check:** after replacement, no `[...]` placeholder may remain. If one does, flag the field.
- **Output:** the draft note with real names, ready for the doctor.

C6. Doctor review screen

- **Layout:** transcript on the left, editable form on the right. Flagged fields are highlighted in amber with the reason shown.
- **Actions:** edit any field; **Approve** (saves as final); **Regenerate** (runs C2–C5 again).
- **Rule:** Approve stays disabled while any required field is empty.

C7. Storage

- **Job:** save the AI draft, the doctor's final version, and who approved it.
- **Output:** an approved note in the database. This approved note is what Modules 2–6 will read later.

The form

The note has 13 fields. Identity fields come from the transcript header, not the LLM, so the AI never handles real names. "Required" means Approve stays disabled until the field is filled.

Field	Type	Filled by	Required	Example
transcript_id	text	System (header)	Yes	T001
patient_id	text	System (header)	Yes	MF-00001
patient_name	text	System (header)	Yes	Ama Mensah
consultation_date	date	System (header)	Yes	2026-10-07
chief_complaint	text	LLM	Yes	Fever and headache for 3 days
symptoms	list of text	LLM	Yes	fever, headache, body pains, loss of appetite
vital_signs	temperature_c, blood_pressure, pulse_bpm, weight_kg	LLM	No (null if not said)	38.9, 120/80, 96, 62
exam_findings	text	LLM	No	Pale conjunctiva, no neck stiffness
diagnosis	text	LLM	Yes	Uncomplicated malaria
diagnosis_code	ICD-10 code	LLM	Yes	B54
medications	list of {name, dose, frequency, duration_days}	LLM	No	Artemether-lumefantrine, 4 tablets, twice daily, 3

Field	Type	Filled by	Required	Example
follow_up_date	date	LLM (from "come back in X days")	No	2026-10-14
flags	list of {field, reason}	System (validation)	n/a	{diagnosis_code, "not a valid ICD-10 format"}

The LLM prompt

Store this as `prompts/scribe_v1.txt`. Every saved note records which prompt version produced it, so changes can be compared later.

```
You are a clinical documentation assistant for an outpatient clinic in Ghana.
You read a doctor-patient consultation transcript and fill a structured
clinical note using the provided tool.

Rules:
1. Use only information stated in the transcript. Never guess or add facts.
2. If a field is not mentioned in the transcript, return null for it.
3. Keep placeholders such as [PATIENT_1] or [PHONE_1] exactly as written.
4. diagnosis: use the doctor's final diagnosis, not the patient's guess.
5. diagnosis_code: the ICD-10 code that best matches that diagnosis.
6. medications: one entry per drug, with name, dose, frequency and
   duration_days, exactly as the doctor prescribed.
7. follow_up_date: if the doctor says "come back in X days/weeks",
   add that to the consultation date. Otherwise null.
8. The transcript may contain Twi or Pidgin. Write the note in English.

Consultation date: {consultation_date}

Transcript:
{masked_transcript}
```

The form fields from *The form* (the LLM-filled ones only) are passed as the tool's JSON schema, so the model cannot answer in free text.

Guardrails

The rule behind every guardrail: a bad AI answer may slow the doctor down, but it can never block them or get saved without their approval.

Validation checks (C4)

Check	Rule	If it fails
Schema	Every field present with the right type	Retry once
Required fields	chief_complaint, symptoms, diagnosis, diagnosis_code not empty	Retry once
Code format	diagnosis_code matches `^[A-Z][0-9]{2}(\.[0-9]{1,2})?$`	Flag field
Medications	Each entry has name, dose, frequency, duration_days	Flag field
Follow-up date	Between consultation date and 90 days after	Flag field
Vital signs range	Temperature 34–43 °C, pulse 30–220 bpm	Flag field
Leftover placeholders (after C5)	No `[...]` text remains	Flag field

Bad response handling

1. **Broken or incomplete answer:** retry once, adding the error message to the prompt ("diagnosis_code was missing, please fill it").
2. **Still broken after the retry:** stop retrying. Send the draft to the doctor with status `needs_attention` and the failed fields highlighted.
3. **API error or timeout (30 seconds):** retry twice, waiting 2 then 4 seconds. If it still fails, open the review screen with an empty form and the banner "AI unavailable, fill manually".
4. **Nothing is ever saved as final without the doctor clicking Approve.**

Storage, statuses and audit trail

One table, `clinical_notes`, keeps both the AI's draft and the doctor's final version, so we can measure how much the doctor had to change.

Column	What it holds
id	Note ID
transcript_id, patient_id	Links to the source

Column	What it holds
ai_draft	The form as the AI filled it (JSON, after unmasking)
final_note	The form after doctor edits (JSON)
edited_fields	List of fields the doctor changed
flags	Flags raised by validation
status	`ready_for_review`, `needs_attention` or `approved`
model, prompt_version	e.g. claude model name, `scribe_v1`
created_at, approved_at, approved_by	Timestamps and reviewer name

Status flow: a new draft is `ready_for_review` (all checks passed) or `needs_attention` (something flagged). Either becomes `approved` only when the doctor clicks Approve.

The name map from C2 is never written to this table; it is held only while the note is being processed.

Tech for the prototype

Part	Choice
Backend	Python, FastAPI, Pydantic (form schema + validation)
LLM	Anthropic SDK, Claude by default; model name set in `.env` so it can be swapped
Database	PostgreSQL (SQLite is fine for day one)
Front end	React page with the review screen
Run	Docker Compose, one command to start

API endpoints

- `GET /transcripts`: list the 10 transcripts.
- `POST /notes/draft` with a transcript_id: runs C2–C5 and returns the draft with its status and flags.
- `PUT /notes/{id}`: saves the doctor's edits.
- `POST /notes/{id}/approve`: marks it approved.

Definition of done

Module 1 is done when all of these work on screen:

- [ ] All 10 transcripts load from the dropdown
- [ ] The masked text sent to the LLM contains no real name, phone or NHIS number (log it to check)
- [ ] Each transcript produces a filled form in under 30 seconds
- [ ] Real names appear correctly on the review screen
- [ ] A broken field shows in amber with its reason
- [ ] The doctor can edit, approve, and see the status change to approved
- [ ] The database shows both the AI draft and the final note, with edited fields listed
- [ ] AI output matches the answer key on diagnosis and diagnosis code for at least 8 of 10 transcripts

Synthetic data prompt

Ten transcripts are enough for the prototype. Each one is generated from an answer key first, so we can score the AI's note against it. Give this prompt to your coding agent:

```
Create 10 synthetic outpatient consultation files for a Ghanaian clinic.
Save each as data/transcripts/T001.json to T010.json.

For each file, FIRST write the answer key, THEN write the transcript from it.

Each file is one JSON object with three parts:

1. header:
   transcript_id (T001...), patient_id (MF-00001...),
   patient_name (fake Ghanaian name), phone (fake, format 024 XXX XXXX),
   nhis_number (fake, format NHIS-XXXXXXXX), consultation_date (in 2026),
   clinic (fake clinic name in a Ghanaian town)

2. answer_key (exactly these fields):
   chief_complaint, symptoms (list), vital_signs (temperature_c,
   blood_pressure, pulse_bpm, weight_kg; use null for any not measured),
   exam_findings, diagnosis, diagnosis_code (ICD-10),
   medications (list of name, dose, frequency, duration_days),
   follow_up_date (or null)

3. transcript: doctor-patient dialogue, 20-40 turns, each line starting
   DOCTOR: or PATIENT:
```

```
One diagnosis per file:
  T001 uncomplicated malaria, T002 hypertension, T003 type 2 diabetes,
  T004 upper respiratory infection, T005 typhoid fever,
  T006 gastroenteritis, T007 urinary tract infection,
  T008 routine antenatal visit, T009 skin infection, T010 asthma

Realism rules:
- The transcript must contain every fact in the answer key, said
  naturally. Never say the ICD-10 code out loud.
- In 3 files the doctor confirms the patient's phone or NHIS number
  out loud (this tests masking).
- In 2 files the patient uses some Twi or Pidgin phrases.
- In 2 files the patient is vague or corrects themselves.
- In 1 file no vital signs are taken (tests null handling).
- In 1 file the doctor mentions another person by name, e.g. a
  relative (tests masking of other names).
- All names, phone numbers and IDs are fake.

After generating, print a table: transcript_id, diagnosis,
number of turns, which test cases it covers.
```

When it's done, read 3 of the files yourself. Check that each conversation actually supports its diagnosis, and that the doctors don't all sound the same.